

LEVEL

1

FOUNDATION (PYTHON + API + GIT + n8n INTRO)

Strong Foundation = Strong Future 🚀

Python Basics • API Basics • Git Basics • n8n Introduction • Mini Projects

LEVEL GOAL



Python, API, Git आणि n8n चे मूलभूत ज्ञान घेऊन छोटे projects बनवणे आणि automation ची सुरुवात करणे.



DURATION

No fixed time
(Skill-based)



WEEKLY HOURS

6 - 8 Hours
(Flexible)



DIFFICULTY

Easy
(Beginner Friendly)



IMPORTANCE

Very High
(Everything Starts Here)



NEXT LEVEL

Level 2
Build (Backend)

1. WHAT YOU WILL LEARN

- ✓ Python Core Basics
Variables, Data Types, Operators
- ✓ Control Flow
if-else, loops, break, continue
- ✓ Functions
Def, return, arguments, scope
- ✓ Error Handling
try, except, finally
- ✓ File Handling
Read/Write JSON, CSV
- ✓ Data Structures
List, Dict, Tuple, Set
- ✓ List Comprehensions
Clean & efficient code
- ✓ Git Basics
Commit, Branch, Push, Pull
- ✓ API Basics
REST, requests, JSON parsing
- ✓ Environment Variables
.env file, keep keys secret
- NEW** n8n Intro (Parallel)
Install, Basic Workflow (Trigger → Action)



2. TOPICS TO MASTER

- 1 Python Setup (VS Code, Pip, venv)
- 2 Variables & Data Types
- 3 Operators & Expressions
- 4 Control Flow (if-else, loops)
- 5 Functions & Scope
- 6 Error Handling (try/except)
- 7 File Handling (JSON, CSV)
- 8 Lists, Dictionaries, Tuples, Sets
- 9 List Comprehensions
- 10 Git Basics (Local + GitHub)
- 11 API Basics (GET, POST, JSON)
- 12 n8n Basics (Trigger → Action)

PARALLEL GROWTH (STARTS HERE)



3. PRACTICAL PROJECTS

1 Python CLI Project

Currency Converter
OR
To-Do File Manager



2 API Data Fetcher

Fetch data from a free public API
(Weather / News / Joke)



3 n8n Mini Automation

RSS Feed → Google Sheet
(Trigger → Action)



WHY PROJECTS FROM DAY 1?

- ✓ Hands-on learning
- ✓ Break, Debug, Fix
- ✓ Build real confidence
- ✓ Portfolio starts early



4. ESSENTIAL TOOLS



VS Code



Python 3.x



Git & GitHub



Postman / Thunder Client



Requests Library



Python-Dotenv



n8n (Self-hosted / Cloud)



Google Sheets

5. CHECKPOINT (Before Moving to Level 2)

- ✓ स्वतः Python script लिहून run आणि debug करता येते.
- ✓ REST API चे documentation वाचून data fetch करून JSON parse करता येते.
- ✓ GitHub वर repo create करून code push करता येतो.
- ✓ n8n मध्ये basic workflow (trigger-action) बनवता येतो.

7. COMMON MISTAKES TO AVOID

- ✗ Documentation न वाचता copy-paste करणे.
- ✗ Errors आल्यावर लगेच हार मानणे.
- ✗ Git commit messages अपस्पष्ट ठेवणे.
- ✗ Hardcoded API keys वापरणे (बेहमी .env वापरा).
- ✗ Projects skip करून फक्त theory शिकत बसणे.



6. PYTHON DEPTH GUIDE (For Full Roadmap)

DEEP (Must Master)

Variables, Control Flow, Functions, Try/Except, File Handling (JSON/CSV), List Comprehensions, Dicts, Requests, JSON Module

STRONG (Later Levels)

OOP Basics, Asyncio, Logging, Dotenv, Pydantic, Pytest, Type Hints, Context Managers

MEDIUM (Use & Understand)

Generators, Dataclasses, Basic Threading/ Multiprocessing, Regex

SKIP (Not Needed)

Metaclasses, CPython Internals, C Extensions, Complex Metaprogramming

Key Libraries Ahead: pydantic, sqlalchemy, fastapi, openai/anthropic SDK, langgraph/langchain, pytest, python-dotenv, asyncio

8. HOW THIS LEVEL BUILDS THE FOUNDATION



Python Basics



API Basics
(JSON, REST)



Git & GitHub
(Code Management)



n8n Intro
(Automation Start)



Combine All
in Mini Projects



Ready for
Level 2 (Build)

SUCCESS FORMULA

Consistency + Practice + Projects = Confidence
Small Steps Daily → Big Results Later



MINDSET

- ✓ Don't just learn. Build.
- ✓ Break things, Debug, Fix, Improve.
- ✓ Document everything you learn.

NEXT LEVEL PREVIEW

In Level 2, आपण स्वतःचा Backend बनवणार, Database जोडणार, Testing शिकणार, Docker वापरणार आणि Cloud वर deploy करणार.



★ FOCUS TODAY, FUTURE WILL BE AMAZING! 🚀

ONE SKILL AT A TIME. MASTER IT. MOVE AHEAD. 🔥

LEVEL
2

BUILD

(BACKEND + SQL + TESTING +
CLOUD BASICS + DOCKER + n8n WEBHOOKS)

Build Your Backend. Connect Everything. 🚀

Python

FastAPI

SQL

Docker

Pytest

n8n

Cloud Deploy

LEVEL GOAL

🎯

Database connected
REST API बनवून live deploy
करणे आणि n8n सोबत
एकत्र जोडणे.

📅

DURATION

No fixed time
(Skill-based)

🕒

WEEKLY HOURS

8 - 12 Hours
(Flexible)

📊

DIFFICULTY

Medium
(Keep Practicing)

★

IMPORTANCE

Very High
(Build Real Systems)

➡️

NEXT LEVEL

Level 3
AI Vocabulary

📖 1. WHAT YOU WILL LEARN

✓ SQL Basics
SELECT, WHERE, JOIN,
GROUP BY, ORDER BY

✓ Database Design
Tables, Relations, Primary Key,
Foreign Key

✓ FastAPI Fundamentals
Routes, Path Params,
Query Params

✓ Pydantic Models
Request/Response Validation

✓ SQLAlchemy Basics
ORM, CRUD Operations

✓ Error Handling
HTTP Exceptions, Validation

✓ Testing Basics
pytest, Unit Tests,
API Tests, Mocking

✓ Docker Basics
Dockerfile, Build, Run Container

✓ Cloud Deploy (Basics)
Render / Railway / AWS Free Tier

✓ n8n Webhooks & HTTP
Webhook Trigger, HTTP Request
Node, Call Your API

2. TOPICS TO MASTER

1 SQL: CRUD Operations

2 SQL: JOINS (INNER, LEFT, RIGHT)

3 SQL: Aggregation & GROUP BY

4 Database Design: 1NF to 3NF

5 FastAPI: Setup & First App

6 FastAPI: Path & Query Params

7 FastAPI: Pydantic Models

8 FastAPI: Dependency Injection

9 FastAPI: Error Handling

10 SQLAlchemy: Setup & Models

11 SQLAlchemy: CRUD with DB

12 Environment Variables (.env)

13 Testing: pytest Basics

14 Testing: API Tests with TestClient

15 Mocking External Calls

16 Docker: Dockerfile Basics

17 Docker: Build & Run Container

18 Deploy: App Live on Free Tier

19 n8n: Webhook Trigger

20 n8n: HTTP Request Node

21 n8n: Call Your FastAPI

3. PRACTICAL PROJECTS

1 Notes API
CRUD API with
User & Notes
(FastAPI + SQLAlchemy)

2 Expense Tracker API
Categories, Expenses,
Summary API
(FastAPI + SQLAlchemy)

3 Deploy Your API
Deploy on Render /
Railway / AWS Free Tier

4 n8n Integration
n8n webhook hits your
API → Add/Update data
→ Get response

WHY THIS LEVEL IS IMPORTANT?

✓ Backend हा तुम्हा सर्व AI/Agent systems
चा बळक आहे.

✓ API सिवाय कुवलीही integration
शक्य नाही.

✓ Production-ready code लिहिण्याची सवय
लागते (Testing + Deploy).

4. TOOLS YOU WILL USE

VS Code

Python 3.x

FastAPI

Pydantic

SQLA SQLAlchemy

PostgreSQL / SQLite

Pytest

Docker Desktop

Render / Railway

n8n

Postman

Git & GitHub

5. CHECKPOINT (Before Moving to Level 3)

✓ Database design करून tables create करता येतात.

✓ FastAPI मध्ये पूर्ण CRUD API बनवून चालवता येते.

✓ SQLAlchemy वापरून DB ऑपरेट करता येते.

✓ pytest ने unit test आणि API test लिहिता येतात.

✓ Docker मध्ये app build आणि run करता येते.

✓ Free-tier पर app live deploy केलेले आहे.

✓ n8n मधून webhook trigger होऊन तुम्हा API ला call करता येते.

6. PYTHON DEPTH GUIDE (For This Level & Beyond)

DEEP (Must Master)

FUNCTIONS, Error Handling, File Handling, JSON, CSV, Dicts,
List Comprehensions, requests, json, os, datetime

STRONG (Solid)

OOP Basics, logging, dotenv, pydantic, pytest,
type hints, context managers, sqlalchemy basics, asyncio

MEDIUM (Understand & Use)

Generators, dataclasses, basic threading/multiprocessing, regex

SKIP (Not Needed)

Metaclasses, CPython internals, C extensions,
Complex Metaprogramming

Key Libraries Now: pydantic, fastapi, sqlalchemy, pytest, requests, python-dotenv

7. COMMON MISTAKES TO AVOID

✗ Hardcoded secrets / API keys.

✗ No input validation → API crashes.

✗ Skipping tests (नेतर मोठा त्रास होतो).

✗ DB relations न समजता tables बनवणे.

✗ Docker मध्ये .env copy करणे.

8. HOW THIS LEVEL CONNECTS (DEPENDENCY FLOW)

Python & API → SQL & DB → FastAPI Backend → Testing → Docker → Cloud Deploy → n8n WebHooks

हे सगळे मिळून तुम्हा Production-ready Backend तयार होतो → पुढे AI & Agents जोडता येतात.

9. WEEKLY STRATEGY (Example)

Mon: SQL + DB

Tue: FastAPI Basics

Wed: SQLAlchemy

Thu: Testing (pytest)

Fri: Docker Basics

Sat: Deploy Live

Sun: n8n Integration + Project

Theory कमी, Code जास्त, Project Daily!

10. MINI PROJECT IDEAS (Extra Practice)

• Bookmark Manager API

• Task Manager API

• Contact Manager API

• Invoice Generator API

• URL Shortener API

11. NEXT LEVEL PREVIEW (Level 3)

AI Vocabulary: Probability, Vectors, Embeddings,
Precision/Recall, Overfitting — फक्त Concept,
Math नाही.

★ FOCUS: BUILD A STRONG BACKEND FOUNDATION. | CLEAN CODE + TESTS + DEPLOY = CONFIDENCE 🚀

LEVEL 3

AI VOCABULARY (Light Math / ML)

2-3 WEEKS ONLY

Conceptual Understanding, Not Calculation.

Build Intuition for AI/ML World

LEVEL GOAL



AI / ML related articles, docs
आणि videos समजून घेण्यासाठी
लागणारी conceptual clarity
मिळवणे.



DURATION

2 - 3 Weeks
(Skill-based)



WEEKLY HOURS

5 - 8 Hours
(Flexible)



DIFFICULTY

Easy to Medium
(Conceptual)



IMPORTANCE

High
(Foundation for AI)



NEXT LEVEL

Level 4
LLM Core

1. WHAT YOU WILL LEARN

- ✓ **Probability & Statistics (Basic)**
Probability, Mean, Median, Variance, Distribution (intuition)
- ✓ **Vector & Dot Product Intuition**
Vectors म्हणजे काय, Dot Product म्हणजे काय, Cosine Similarity concept समजून घेणे
- ✓ **Evaluation Metrics**
Precision, Recall, F1-score, Accuracy - कोणत्या situations मध्ये काय वापरायचे ते समजणे
- ✓ **Supervised Learning Concepts**
Labels, Features, Dataset, Training vs Testing
- ✓ **Overfitting vs Underfitting**
सोप्या भाषेत समजून घेणे
- ✓ **Neural Networks (Concept)**
Neurons, Layers, Activation Functions - थोडा idea level
- ✓ **Transformer (High Level)**
Attention म्हणजे काय, Transformer architecture म्हणजे काय
- ✓ **Fine-tuning (Concept)**
कधी आणि का वापरतात
- ✓ **Quantization (Concept)**
Model सहान / जलद का करतात

REMEMBER

Calculate करण्याची गरज नाही.
समजून घेणे हीच या level ची purpose आहे.

2. TOPICS EXPLAINED (CONCEPT ONLY)

- 1 Probability**
Event, Sample Space, Probability (0 to 1), Independent Events
- 2 Statistics**
Mean, Median, Mode, Variance, Std Deviation - intuition
- 3 Vectors (Intuition)**
Vector म्हणजे number ची list, Direction आणि Magnitude
- 4 Dot Product**
दोन vectors चा similarity measure कसा काढतात - concept
- 5 Cosine Similarity**
Angle आधारित similarity, 0 ते 1 पर्यंत value
- 6 Evaluation Metrics**
Precision vs Recall trade-off, F1-score balanced measure
- 7 Supervised Learning**
Training data, Labels, Model, Prediction
- 8 Overfitting**
Model training data कुपु चांगले शिकते, पण new data वर खराब perform करते
- 9 Key AI Concepts (Extra Awareness)**
 - ✓ Neural Network म्हणजे काय?
 - ✓ Transformer म्हणजे काय?
 - ✓ Attention म्हणजे काय?
 - ✓ Fine-tuning म्हणजे काय आणि कधी?
 - ✓ Quantization म्हणजे काय?

3. HOW TO STUDY (EASY APPROACH)

- Read / Watch**
Short notes, visuals, videos (concept level)
- Understand in Simple Words**
सक्या सध्या लिख किंवा explain करा
- Use Real Examples**
Daily life / simple examples सोबत जोडा
- Don't Calculate**
Formula memorize / calculation यात वेळ घालू नका
- Make Your Own Summary**
प्रत्येक topic चे 1 page summary तयार करा
- Test Yourself**
Concepts कुणाला तरी explain करा किंवा खुबद लिखून बघा

TIME LIMIT RULE

या Level वर 2-3 आठवड्यांचे
जास्त वेळ घालू नका.
Keep it light and move ahead!

4. MINI PRACTICE (NO CALCULATION)

- ✓ Coin toss experiment (probability intuition)
- ✓ Student marks example (mean, median concept)
- ✓ Movie recommendation example (cosine similarity idea)
- ✓ Simple classification example (spam vs not spam)
- ✓ Overfitting example (memorization vs generalization)

5. WHAT NOT TO STUDY (SKIP)

- ✗ Linear Algebra Proofs (Matrix, Eigen values, etc.)
- ✗ Calculus (Derivatives, Integrals, Limits)
- ✗ Backpropagation Math
- ✗ Deep Statistics
- ✗ PyTorch / TensorFlow Model Training
- ✗ Advanced ML Algorithms (SVM, XGBoost, etc.)

6. CHECKPOINT (Before Moving to Level 4)

- ✓ Probability, mean, variance सारखे संकद समजतात.
- ✓ Vectors, dot product, cosine similarity चा concept समजतो.
- ✓ Precision, Recall, F1-score कसासाठी वापरतात हे समजते.
- ✓ Overfitting म्हणजे काय हे आपल्याला कळते.
- ✓ Neural Network, Transformer, Attention हे concepts समजून सांगता येतात.
- ✓ AI problem बदल वाढत्यावर धाबदार वाटत नाही.



7. WHY THIS LEVEL IS IMPORTANT?

- ★ AI / ML जगाची भाषा समजायला सुरुवात होते.
- ★ RAG, Embeddings, Agents सासाठी ही conceptual पायाभरणी आहे.
- ★ Interview आणि articles समजणे confidence मिळते.
- ★ Time वाचतो - Deep math मध्ये अडकत नाही.



8. KEY TAKEAWAY

- ✓ तू Data Scientist नाही, तू Applied AI Engineer आहेस.
- ✓ Concept समजून घे, System build कर, Value deliver कर.

9. DEPENDENCY FLOW (Why This Level is Needed)



10. WEEKLY STUDY PLAN (EXAMPLE)

- | | |
|-------------------------------|---|
| Day 1 : Probability Basics | Day 6 : Supervised Learning Concepts |
| Day 2 : Statistics Basics | Day 7 : Overfitting |
| Day 3 : Vectors & Dot Product | Day 8 : Neural Network Concept |
| Day 4 : Cosine Similarity | Day 9 : Transformer & Attention |
| Day 5 : Evaluation Metrics | Day 10 : Fine-tuning & Quantization |
| | Day 11+: Revision + Summary + Mini Test |

11. RESOURCES (CONCEPT LEVEL)

- 3Blue1Brown (Essence of Linear Algebra playlist - watch first 3-4 videos only)
- StatQuest with Josh Starmer (Probability, Statistics, ML Concepts)
- Google Machine Learning Crash Course - Concepts (Read only)
- Medium / Towards Data Science - Beginner Articles

12. NEXT LEVEL PREVIEW (LEVEL 4)

LLM कसे काम करता ते शिकणार -
Tokens, Context Window, API, Function Calling, Structured Output, Local LLMs आणि Evaluation.
हा Level समजला की AI apps बनवण्याला खरी सुरुवात होते!



★ FOCUS ON CONCEPTS, NOT CALCULATIONS.

| LIGHT LEARN. CLEAR UNDERSTAND. FAST BUILD.

| KEEP MOVING FORWARD!



GOLDEN RULE: 2-3 WEEKS MAX. CONCEPTS समजले की लगेच पुढे जा आणि BUILD करा!

ONE SKILL AT A TIME. MASTER IT.

LEVEL
4

LLM ENGINEERING CORE

(Tokens, APIs, Function Calling, Context Engineering)

Work with LLMs Like a Pro. Build Reliable AI Systems. 🚀

Python APIs n8n Prompt Engineering Local LLM (Ollama) Evaluation

LEVEL GOAL



LLMs समजून वापरायचे, योग्य context द्यायचा, structured output द्यायचा आणि production-ready AI features build करायचे.



DURATION

No fixed time
(Skill-based)



WEEKLY HOURS

8 - 12 Hours
(Flexible)



DIFFICULTY

Medium
(Keep Practicing)



IMPORTANCE

Very High
(Core of AI Engineer)



NEXT LEVEL

Level 5
RAG (Vector Search)

1. WHAT YOU WILL LEARN

- ✓ LLM Fundamentals
Tokens, Context Window, Temperature, Top-p, Top-k
- ✓ OpenAI / Anthropic APIs
Chat Completions, Streaming, System & User Messages
- ✓ Prompt Engineering (Basics)
System Prompts, Few-shot, Chain-of-Thought
- ✓ Function / Tool Calling
Define Tools, Call Functions, Handle Responses
- ✓ Structured Output (JSON Mode)
Get Clean JSON, Pydantic Models
- ✓ Context Engineering (Core)
Right Info, Right Time, Right Format in Context
- ✓ Model Selection & Routing
Cost vs Quality, Task Complexity, Smart Model Routing
- ✓ Semantic Caching (Cost Save)
Cache Similar Queries & Responses
- ✓ Local LLMs with Ollama
Run Llama / Mistral Locally (Test, Compare, Zero Cost)
- ✓ n8n with AI
Use LLM Node in n8n, Simple AI Automations
- ✓ Basic Evaluation (Start Here)
Correctness, Coherence, LLM-as-Judge Intro



2. TOPICS TO MASTER

- 1 LLM Basics: Tokens, Context, Costs
- 2 Temperature, Top-p, Top-k (Sampling)
- 3 OpenAI / Anthropic API Setup
- 4 Chat Completion (Non-Streaming)
- 5 Streaming Responses in Python (SSE)
- 6 System Prompt & Role of Messages
- 7 Few-shot Prompting
- 8 Chain-of-Thought Prompting
- 9 Function / Tool Calling (Basics)
- 10 Advanced Tool Calling (Multiple Tools)
- 11 Structured Output (JSON Mode)
- 12 Pydantic Models with LLM Output
- 13 Context Engineering Principles
- 14 Model Selection & Routing Strategy
- 15 Semantic Caching (Redis / Simple Cache)

PARALLEL GROWTH (CONTINUES)

Python + APIs + n8n + AI (All Together)
Projects run in parallel with learning

3. PRACTICAL PROJECTS

1 AI Summarizer API

Long text summary with streaming response



2 Function Calling Demo

User asks → LLM calls function → get result → reply



3 Structured Data Extractor

Extract data from text in clean JSON format



4 n8n AI Automation

Email आता → AI summary → Slack ला पादवणे



WHY PROJECTS MATTER?

- ✓ Concepts कळत
- ✓ Confidence वाढते
- ✓ Real-world thinking येते
- ✓ Portfolio लवकर तयार होते



4. TOOLS & TECHNOLOGIES



VS Code



Python 3.x



OpenAI API / Anthropic API



Ollama (Local LLM)



n8n



Pydantic



Requests / httpx



python-dotenv



Postman



LangChain (Optional)



Redis (For Caching)

5. CHECKPOINT (Before Moving to Level 5)

- ✓ सुरत्याही LLM API usage ने text generate करता येतो.
- ✓ Streaming response real-time मध्ये handle करता येते.
- ✓ Function / Tool calling वापरून structured data मिळते.
- ✓ JSON mode आणि Pydantic validations समजता.
- ✓ योग्य context कसा द्यायचा हे समजत आणि उपयोगी करता येतो.
- ✓ n8n मध्ये AI node वापरून automation बनवता येतो.
- ✓ साधे evaluation checks करून output quality चेक करता येते.



6. PYTHON DEPTH GUIDE (For This Level & Beyond)

DEEP (Must Master)

Functions, Error Handling, File Handling (JSON), requests/httpx, json, os, datetime, try/except

STRONG (Solid)

pydantic, python-dotenv, logging, type hints, asyncio (basic), context managers

MEDIUM (Understand & Use)

Generators, dataclasses, regex, basic threading/concurrency

SKIP (Not Needed)

Metaclasses, CPython internals, C extensions, Complex Metaprogramming

Key Libraries Added: openai / anthropic SDK, pydantic, langchain (basic)

7. COMMON MISTAKES

- ✗ Prompt थूप सोब करून irrelevant context देणे.
- ✗ Temperature जास्त ठेवून inconsistent outputs येणे.
- ✗ Function schema चुकीचे बनवणे.
- ✗ JSON mode न वापरता messy output accept करणे.
- ✗ Cost / token usage monitor न करणे.



8. HOW THIS LEVEL BUILDS THE FOUNDATION FOR NEXT LEVELS



LLM Core

LLMs सोबत काम करायचा सिकलो.



Structured Output (JSON + Tools)

RAG आणि Agents साठी डेटा स्पष्ट मिळतो.



Context Engineering Principles

RAG retrieval आणि Agents मध्ये accuracy वाढते.



Model Selection & Routing

Cost control आणि performance optimize करता येते.



Evaluation Basics

RAG, Agents आणि Production साठी quality check करता येते.



Ready for Level 5 (RAG)

Embeddings + Vector Search साठी तयारी पूर्ण!

9. WEEKLY STRATEGY (Example)

- | | |
|---------------------------------------|----------------------------------|
| Day 1 : LLM Basics + Tokens + Context | Day 6 : Structured Output (JSON) |
| Day 2 : API Setup + Chat Completion | Day 7 : Context Engineering |
| Day 3 : Streaming Response | Day 8 : Model Routing + Caching |
| Day 4 : Prompt Engineering | Day 9 : Local LLM with Ollama |
| Day 5 : Function Calling Basics | Day 10 : n8n + AI Automation |

🔑 Theory करी, Code जास्त, Project Daily!

10. MINI PROJECT IDEAS (Extra Practice)

- AI Blog Summarizer
- Resume Data Extractor (JSON Output)
- AI Email Responder (Draft)
- YouTube Video Summarizer
- Product Review Analyzer

11. NEXT LEVEL PREVIEW (LEVEL 5)

Embeddings म्हणजे काय?
Vector Database म्हणजे काय?
Documents ला chunk करून search कसा करावा?
RAG Pipeline कसा बनवायचा?
Evaluation (Faithfulness, Relevance) basics!



★ REMEMBER: Right Context + Right Model + Right Tools = Powerful AI System | 🏆 Build. Test. Improve. Repeat. 🚀

LEVEL 5

RAG (Retrieval Augmented Generation)

Find Right Information. Give Accurate Answers. 🎯

🔗 Embeddings 🗄️ Vector DB 🔍 Hybrid Search 📄 Chunking 🔄 Reranking 📊 Evaluation

LEVEL GOAL



Complete RAG pipeline build करून high-quality Q&A system तयार करणे आणि त्याची quality evaluate करणे. RAG कधी वापरायचा आणि कधी नाही हे समजणे.



DURATION

No fixed time
(Skill-based)



WEEKLY HOURS

8 - 12 Hours
(Flexible)



DIFFICULTY

Medium
(Practice Heavy)



IMPORTANCE

Very High
(Core for AI Apps)



NEXT LEVEL

Level 6
Agents
(Tools + Memory + MCP)

1. WHAT YOU WILL LEARN

- What is RAG & When to Use
Use cases, pros, cons
- Embeddings Basics
Dense vectors, meaning, similarity search
- Embedding Models
OpenAI, Sentence-Transformers, bge, all-MiniLM
- Vector Databases
Chroma, Qdrant, FAISS (Concept + Basic Use)
- Document Loading
PDF, TXT, CSV, Web, Notion, API
- Chunking Strategies
Fixed, Recursive, Semantic, Sliding Window
- Metadata & Filtering
Metadata store, Filtering, Tag based search
- Hybrid Search
Vector + BM25 (Keyword) combined search
- Reranking
Cross-Encoder, Cohere Rerank, Top-K reranking
- Query Rewriting
Rewrite for better retrieval
- Multi-Query Retrieval
Generate multiple queries, expand results
- Context Compression
Reduce tokens, extract important parts
- Grounded Generation
Answer with sources / citations, avoid hallucination
- Basic RAG Evaluation
Faithfulness, Relevance, Context Recall



2. TOPICS TO MASTER

- RAG Overview & Architecture
- Embeddings Concept
- Embedding Models Comparison
- Vector DB Setup (Chroma/Qdrant)
- Load Documents & Data Sources
- Chunking Techniques
- Store & Index Embeddings
- Similarity Search (Top-K)
- Metadata & Filtering
- Hybrid Search (BM25 + Vector)
- Query Rewriting (LLM based)
- Multi-Query Retrieval
- Reranking Retrieved Results
- Context Compression
- Grounded Generation (Answer + Source)
- RAG Evaluation Basics
- When NOT to use RAG

RAG PIPELINE (BIG PICTURE)



3. PRACTICAL PROJECT

RAG Q&A System (Document Based Chatbot)



- Ingest Documents
PDF / TXT / Web pages
- Chunk, Embed & Store
Store in Vector DB
- Retrieve Relevant Chunks
Hybrid Search + Filters
- Rerank Results
Get best relevant context
- Generate Answer
LLM with context
- Show Sources
Citations with file + page

EXTRA FEATURES (Nice to Have)

- Chat History Memory
- Metadata Filters (date, tag, type)
- Source Highlight in Answer
- Export Chat / Share Link
- Feedback (Good / Bad Answer)
- Simple Evaluation Report

WHY THIS LEVEL IS IMPORTANT?

- Real-world AI apps ची पहिली पायरी
- Knowledge + LLM एकात्र काम करतात
- Hallucination कमी करण्यासाठी महत्वाचे
- Agents साठी foundation तयार होते

4. TOOLS & TECHNOLOGIES

- Python 3.x
- LangChain / LangGraph
- Chroma / Qdrant
- FAISS (Optional)
- Sentence-Transformers
- OpenAI / Anthropic API
- Cohere Rerank (Optional)
- n8n
- Streamlit (UI)
- PyPDF / Unstructured
- tiktoken
- dotenv
- VS Code

5. CHECKPOINT (Before Moving to Level 6)

- Documents ingest करून index करू शकतो.
- Embeddings store आणि similarity search करू शकतो.
- Hybrid search + filters वापरू शकतो.
- Reranking लावून result quality improve करू शकतो.
- Answer सोबत source/citation दाखवू शकतो.
- Basic evaluation करून system सुधारू शकतो.
- RAG कधी वापरायचा आणि कधी नाही हे समजले आहे.



6. COMMON MISTAKES TO AVOID

- खूप मोठे chunks ठेवणे (context waste होतो).
- फक्त vector search (BM25 सोडवाय) वापरणे.
- Reranking skip करणे.
- Source/citation न देणे.
- Hallucination filter / grounding न करणे.
- Evaluation न करता production मध्ये नेणे.



7. WHEN NOT TO USE RAG?

- Real-time changing data (stock price, live score)
- Structured data (database query करा)
- Simple tasks (tool-calling पुरेसे आहे)
- Short term memory / session chat (RAG नको)
- Creative generation (story, image idea etc.)



8. DEPENDENCY FLOW (Why This Level Needs Previous Levels)



9. WEEKLY STRATEGY (Example Path)

- Day 1-2 RAG Overview + Embeddings
- Day 3-4 Chunking + Vector DB Setup
- Day 5-6 Load Documents + Store
- Day 7-8 Retrieval + Hybrid Search
- Day 9 Reranking + Filters
- Day 10 Query Rewriting + Multi Query
- Day 11 Context Compression
- Day 12 Grounded Generation



Practice + Break - Debugs

10. MINI PROJECT IDEAS (Extra Practice)

- Company Docs Q&A Bot
- Research Papers Assistant
- Resume Q&A System
- Product Manual Chatbot
- YouTube Video Q&A (transcript based)
- Personal Knowledge Base Chatbot



11. NEXT LEVEL PREVIEW (LEVEL 6)

पुढील level मध्ये आपण शिकू:

- ReAct Agent, Tools, Memory
- LangGraph, State Management
- MCP (Model Context Protocol)
- Safety, Permissions, Guardrails
- n8n + Agent Integration

एक smart, reliable, secure agent तयार करण्यात आले!



KEY MANTRA:

Right Data + Right Retrieval + Right Context + Right LLM = Accurate & Reliable AI System



LEVEL
6

AGENTS + SECURITY + PROTOCOLS

Build Reliable, Secure & Powerful AI Agents

ReAct Pattern Tools & Functions Memory LangGraph MCP Security n8n + Agents

LEVEL GOAL



Production-ready multi-tool agents बनवणे जे secure, reliable आणि real-world use cases solve करतात. MCP समजून वापरणे, security guardrails सह powerful automations बनवणे.



DURATION

No fixed time
(Skill-based)



WEEKLY HOURS

10 - 15 Hours
(Flexible)



DIFFICULTY

Medium to Hard
(Practice Heavy)



IMPORTANCE

Very High
(Core for AI Engineer)



NEXT LEVEL

Level 7
Automation & Integration Deepen

1. WHAT YOU WILL LEARN

- ✓ ReAct Pattern & Reasoning
Observation → Thought → Action
- ✓ Planning & Task Decomposition
Complex कचे सरान steps मध्ये सोडणे
- ✓ Tools / Function Calling (Advanced)
Dynamic tool selection, parallel calls, error handling
- ✓ Agent Reliability
Retries, fallbacks, timeouts, state consistency
- ✓ State Management
Session state, checkpoints, resumable workflows
- ✓ Short-Term Memory
Conversation history, sliding window, summary memory
- ✓ Long-Term Memory
Vector memory, user profile, persistent storage
- ✓ LangGraph (or similar)
Graph based agent workflows, loops, branches, conditions
- ✓ MCP (Model Context Protocol)
Concept, components, use cases, connectors, security considerations
- ✓ Multi-Tool Agents
Multiple tools orchestration, priority, dependencies
- ✓ Guardrails & Safety
Input / Output guardrails, content moderation
- ✓ Prompt Injection Defense
Attack types, prevention techniques, sandboxing
- ✓ Multi-Modal Integration (Basics)
Vision API (image understanding), Whisper (speech to text)
- ✓ n8n + Agents Integration
Trigger agent from n8n, pass data, handle response, human-in-loop

2. TOPICS TO MASTER

- 1 Agent Types: ReAct, Plan & Execute
- 2 Thought / Action / Observation Loop
- 3 Tool Schema Design (JSON Schema)
- 4 Dynamic Tool Selection (LLM decides)
- 5 Parallel Tool Calls
- 6 Error Handling in Tools
- 7 Human-in-the-Loop Approval
- 8 State & Session Management
- 9 Checkpointing & Resume
- 10 Short-Term vs Long-Term Memory
- 11 Memory Retrieval Strategies
- 12 LangGraph: Nodes, Edges, State
- 13 Conditional Routing in LangGraph
- 14 Loops & Iterative Workflows
- 15 MCP: Architecture & Components
- 16 MCP: Resources, Tools, Prompts
- 17 MCP: Client & Server Example
- 18 Security in MCP (Auth, Permissions)
- 19 Multi-Agent Patterns (Basic)
- 20 Guardrails: Input & Output
- 21 Data Privacy & PII Handling
- 22 Prompt Injection: Types of Attacks
- 23 Defense: Validation, Escaping, Filters
- 24 Multi-Modal: Vision API Basics
- 25 Whisper: Speech to Text
- 26 n8n Agent Trigger (Webhook/Node)
- 27 n8n + Agent Workflow (End-to-End)

AGENT STACK (BUILDING BLOCKS)



3. PRACTICAL PROJECTS

- 1 RAG + Tool Agent
Docs वर Q&A + External tools (Calculator, Search)
- 2 Customer Query Agent
Query समजून tools वापरून answer देणे + human escalation
- 3 Data Analyst Agent
CSV upload → analysis → charts (Python tool + reasoning)
- 4 Personal Assistant Agent
Calendar, Email, Notes, Reminders (via tools)
- 5 n8n + Agent Automation
n8n मध्ये trigger → agent process → response → action

WHY THIS LEVEL IS CRUCIAL?

- ✓ LLM ला real-world कचे करण्याची क्षमता हवे मिळते.
- ✓ Tools + Memory + Planning एकत्र येतात.
- ✓ Security शिवाय production गरूय नाही.
- ✓ पुढच्या सर्व advanced systems ची foundation.

4. TOOLS & TECHNOLOGIES

- Python 3.x
- LangGraph / LangChain
- OpenAI / Anthropic API
- MCP (Claude Ecosystem)
- Pydantic
- Chroma / Qdrant
- Redis (Memory / Cache)
- PostgreSQL / SQLite
- Whisper (OpenAI)
- Vision API (GPT-4o / Claude)
- n8n
- VS Code
- Git & GitHub
- Docker (Basic)

GOOD TO KNOW

MCP अजून universal standard नाही, पण समजून घेतल्यास early advantage मिळते.

5. CHECKPOINT (Before Moving to Level 7)

- ✓ ReAct agent बनवून tools वापरता येतात.
- ✓ Multi-tool agent reliable आहे (errors handle होतात).
- ✓ Short & long-term memory वापरता येते.
- ✓ Human-in-the-loop approval workflow बनवता आहे.
- ✓ Prompt injection attacks पासून protection समजू केला आहे.
- ✓ n8n मधून agent trigger करून end-to-end automation के आहे.



6. SECURITY GROUND RULES

- ✗ High-risk actions (delete, send email, execute, payments) कधीही unlimited permission सह देऊ नका.
- ✗ Permission Scoping: कचेक tool ला मर्यादित access द्या.
- ✗ Human Approval: critical actions साठी approval घ्या.
- ✗ Logging & Audit: सर्व actions log करा.
- ✗ Input Validation: user input validate करा.
- ✗ Data Privacy: PII handle करण्याचे नियम ठेवा.

7. AGENT REALITY CHECK

- ✓ PRODUCTION READY
 - Tool-calling agents
 - RAG agents
 - Human-in-the-loop agents
 - Workflow automation agents
 - ✗ HYPE (NOT REAL YET)
 - Fully autonomous business
 - agenticions.
 - AGI-level agents
- ⚠️ FRAGILE (CARE NEEDED)
- Long-running (10+ steps) agents
 - Large scale multi-agent systems
 - Computer-use agents



8. DEPENDENCY FLOW (Why This Level Needs Previous Levels)



9. WEEKLY STRATEGY (Example Path)

- Day 1-2 : ReAct Pattern + Simple Agent
- Day 3-4 : Tools & Function Calling
- Day 5-6 : Memory (Short + Long Term)
- Day 7-8 : LangGraph Basics
- Day 9-10 : MCP Introduction + Example
- Day 11-12 : Security & Guardrails
- Day 13-14 : n8n + Agent Integration
- Day 15+ : Build & Improve Projects

📌 Theory कमी, Code जास्त, Test जास्त!



10. MINI PROJECT IDEAS (Extra Practice)

- Research Agent (Search + Summarize + Save)
- Travel Planner Agent (Tools + Memory + APIs)
- Meeting Notes Agent (Audio → Transcript → Summary)
- Code Helper Agent (Explain + Run + Debug)
- File Analyst Agent (CSV/PDF → Insights)

Build → Break → Debug → Improve → Repeat



11. NEXT LEVEL PREVIEW (LEVEL 7)

You will learn:

- ✓ Email, Calendar, Slack, CRM Integrations
- ✓ OAuth & Webhooks Deep Dive
- ✓ n8n Advanced Workflows
- ✓ Agent + n8n End-to-End Business Automations
- ✓ ROI & Business Framing

Real-world business automation starts next!



LEVEL
7AUTOMATION + INTEGRATION
(SPECIALIZATION DEEPEN)

Turn AI + Agents into Real-World Business Automations

Email Calendar Slack CRM Webhooks OAuth n8n Advanced AI + Agents

LEVEL GOAL



End-to-end business automation design आणि build करणे, ज्यामुळे manual work कमी होईल, cost वाचेल आणि process smart होईल.



DURATION

No fixed time
(Skill-based)



WEEKLY HOURS

8 - 12 Hours
(Flexible)



DIFFICULTY

Medium to Hard
(Implementation Heavy)



IMPORTANCE

Highest
(Core Specialization)



NEXT LEVEL

Level 8
Production +
System Design

1. WHAT YOU WILL LEARN

- Business Process Understanding
Process mapping, bottlenecks, value analysis
- Email Integration (Deep)
Send, Receive, Reply, Attachments, Templates, Labels
- Calendar Integration
Create events, schedule meetings, reminders, availability
- Slack Integration
Messages, Threads, DM, Files, Slash Commands
- CRM Integration
HubSpot / Zoho / Pipedrive basics, Leads, Contacts, Deals
- Webhooks Deep Dive
Receive data, verify signature, transform & forward
- OAuth 2.0 Basics
Authorization Code Flow, Access Token, Refresh Token
- n8n Advanced Workflows
Multi-branch, loops, split-merge, error handling, sub-workflows
- Agent + n8n Integration
Trigger agent from n8n, use agent output in workflows
- Error Handling & Retries
Fallbacks, notifications, logging failures
- Scheduling & Triggers
Cron, intervals, conditional triggers, polling
- Data Transformation
JSON, CSV, Filters, Code node (JavaScript/Python)
- Security & Secrets in n8n
Credentials, environment variables, permissions



2. TOPICS TO MASTER

- Business Process Mapping
- Email: IMAP, SMTP, Send, Reply
- Email Templates & Variables
- Calendar: Create / Update Events
- Availability Check & Scheduling
- Slack: Messages, Threads, Files
- Slack: Slash Command Integration
- CRM: Leads, Contacts, Deals API
- Webhooks: Receive & Respond
- Webhook Security & Validation
- OAuth 2.0 Flow (Concept + Use)
- n8n: Advanced Nodes Overview
- n8n: IF, Switch, Merge, Loop
- n8n: Sub-workflows & Reusability
- n8n: Error Handling & Retries
- n8n: Data Stores & Variables
- Agent + Workflow Orchestration

AUTOMATION BUILDING BLOCKS



AUTOMATION STACK

n8n + APIs + Agents + Data + Integrations
=
Smart Automation System

3. PRACTICAL PROJECT

AI Lead Qualification & Follow-up System



Lead Capture
Web form / Webhook / Email / CRM



AI Qualification
Agent qualify lead, score, intent detect



CRM Update
Create / Update lead in CRM with score & notes



Slack Notification
Notify team with lead summary & score



Follow-up Email
Send personalized email based on lead type



Calendar Task
Create follow-up task for sales team

BUSINESS IMPACT

- Manual work 70%+ कमी
- Faster response time
- Better lead conversion
- Cost per lead कमी

4. TOOLS & TECHNOLOGIES

- n8n (Advanced)
- Python (Code Node)
- OpenAI / Anthropic API
- LangChain / LangGraph
- Gmail API / SMTP / IMAP
- Google Calendar API
- Slack API
- HubSpot / Zoho / Pipedrive API
- Webhooks
- OAuth 2.0 (Concept)
- JSON / CSV / Airtable
- PostgreSQL / SQLite (Optional)
- JavaScript (Basic for n8n)
- Docker (Optional)
- Environment Variables / .env

5. CHECKPOINT (Before Moving to Level 8)

- कोणताही business process समजून map करता येतो.
- Email, Calendar, Slack, CRM integration स्वतः जोडता येते.
- n8n मध्ये complex multi-step workflow बनवता येतो.
- Agent ला n8n workflow मध्ये integrate करता येतो.
- Error handling, retries आणि logging settings करता येते.
- पूर्वी automation live करून real data वर test करता येते.



6. COMMON MISTAKES

- Process समजून न घेताच automation बनवणे.
- Hardcoding credentials / API keys.
- Error handling न करणे (workflow fail होतो).
- Duplicate data / multiple triggers create करणे.
- Too many steps in single workflow (maintain करणे अवघड).
- Testing न करता production मध्ये push करणे.



7. BUSINESS FRAMING (IMPORTANT)

- Problem काय आहे?
- Solution म्हणून ही automation काय करते?
- Manual steps किती कमी झाले?
- Time / Cost किती वाचते?
- Result / Impact काय मिळतो?



8. DEPENDENCY FLOW (Why This Level Comes After Level 6)



Level 7 मध्ये तु हे सर्व एकत्र वापरून real-world business automation बनवायला मिळतील.

9. WEEKLY STRATEGY (Example)

- | | |
|----------------------------------|---------------------------------|
| Day 1 - Business Process Mapping | Day 8 - Error Handling |
| Day 2 - Email Integration | Day 9 - Agent + n8n Integration |
| Day 3 - Calendar Integration | Day 10 - Build Mini Automation |
| Day 4 - Slack Integration | Day 11 - Test & Debug |
| Day 5 - CRM Integration | Day 12 - Improve & Deploy |
| Day 6 - Webhooks + OAuth Basics | |
| Day 7 - n8n Advanced Nodes | |

Theory कमी, Practice जास्त, Automation Daily!

10. MINI PROJECT IDEAS (Extra Practice)

- Auto Email Responder with AI
- Meeting Scheduler + Reminders
- Slack Ticket to CRM Automation
- Invoice Received → Record + Notify
- Newsletter Automation System
- Support Ticket Triage with AI
- Daily Report Generator & Mailer

11. NEXT LEVEL PREVIEW (LEVEL 8)

तु आता तयार आहेस:

- Logged, evaluated, secure आणि scalable systems बनवायला
- Cost, latency, errors handle करून production-ready बनवायला
- Cloud deploy, monitoring आणि system design समजायला

★ Production World Starts Next!

★ FOCUS: BUSINESSES साठी AUTOMATION. VALUE DELIVER करा. | IMPACT मोजा. | GROW करा.

LEVEL 8

PRODUCTION + EVALUATION + SYSTEM DESIGN

Build Reliable, Observable, Secure & Scalable AI Systems 🚀

Evaluation

Observability

Reliability

Security

Cloud Deploy

System Design

LEVEL GOAL



Production-grade AI applications
are reliable, secure, scalable,
cost-effective and observable. Real users
must live and trustable systems.



DURATION

No fixed time
(Skill-based)



WEEKLY HOURS

10 - 15 Hours
(Flexible)



DIFFICULTY

Medium to Hard
(Implementation Heavy)



IMPORTANCE

Very High
(Career Defining Level)



NEXT LEVEL

Level 9
Portfolio +
Positioning

1. WHAT YOU WILL LEARN

- LLM Evaluation Deep Dive
Correctness, Coherence, Faithfulness, Relevance
- LLM-as-Judge
Golden datasets, Rubrics, Pairwise eval, Grading
- Agent Evaluation
Task completion, Tool selection accuracy, Step accuracy
- Langfuse (Observability)
Traces, Sessions, Scores, Datasets, Experiments
- Prompt Versioning
Prompts as Code, Version control, A/B compare
- Cost & Usage Tracking
Token usage, Cost per request, Cost per workflow
- Latency & Metrics
PSO, P95, Error rate, Throughput basics
- Retries & Backoff
Exponential backoff, Jitter, Max retries
- Timeouts & Rate Limits
Timeout handling
- Fallbacks & Model Routing
Fallback models, Degrade gracefully, Smart routing
- Circuit Breaker Pattern
Stop calling failing service, Recover automatically
- Dead Letter Queue
Failed tasks store, Replay, Reprocess
- Security Hardening
Secrets management, AuthN/AuthZ, Least privilege, Input/Output validation
- Cloud Deployment (Advance)
IAM basics, Env vars, Secrets, Domain, HTTPS, Logging
- Basic System Design
Caching, Queues, Sync vs Async, Scale: 100 users → 1 lakh users

2. TOPICS TO MASTER

- Golden Dataset Creation
- Evaluation Metrics Deep Dive
- LLM-as-Judge (Rubrics + Grading)
- Langfuse Setup & Tracing
- Datasets, Experiments in Langfuse
- Prompt Versioning Strategy
- Token & Cost Monitoring
- Latency Monitoring (PSO/P95)
- Structured Logging (JSON logs)
- Alerts & Notifications (Slack/Email)
- Retry Logic with Exponential Backoff
- Timeout Handling
- Rate Limit Handling
- Fallback Models & Degradation
- Model Routing (Smart Routing)
- Circuit Breaker Pattern
- Dead Letter Queue Pattern
- Secrets Management (.env, Vault)
- Authentication & Authorization
- Cloud Deploy (Render/Railway/AWS)
- System Design Fundamentals
Caching, Queues, DB connection pool, Sync vs Async, Scaling concepts

RELIABILITY PATTERNS



3. PRACTICAL PROJECT

Production Hardened AI System
(Take any project from Level 4-7 and upgrade)

- Observability
Langfuse tracing, metrics, logs, cost tracking
- Evaluation
LLM-as-judge, rubric, eval report
- Reliability
Retry, timeout, fallback, circuit breaker, DLQ
- Security
Auth, secrets, validation, least privilege
- Deployment
Cloud deploy, domain, HTTPS, environment setup
- Monitoring & Alerts
Real-time alerts for errors, latency, cost

PROJECT OUTCOME

- ✓ Live & accessible system
- ✓ Logged, evaluated & monitored
- ✓ Secure & cost-optimized
- ✓ Handles failures gracefully
- ✓ Ready for real users

4. TOOLS & TECHNOLOGIES

- Python 3.x
- Langfuse (Tracing & Eval)
- OpenAI / Anthropic API
- LangChain / LangGraph
- Pydantic
- Redis (Cache / Queue)
- PostgreSQL / SQLite
- Docker
- Render / Railway / AWS (Free Tier)
- Git & GitHub
- Prometheus (Basic)
- Grafana (Basic Dashboards)
- Sentry (Error Tracking)
- JWT / OAuth Basics
- python-dotenv
- VS Code

CLOUD BASICS (FOCUS)

- ✓ IAM (Users, Roles, Policies)
- ✓ Environment Variables & Secrets
- ✓ S3 / Storage Basics
- ✓ Domain + HTTPS (SSL)
- ✓ Logs & Monitoring Setup

5. CHECKPOINT (Before Moving to Level 9)

- System live and stable
- Logs, metrics and traces available
- Evaluation report generate
- Errors handle (retry/fallback/DLQ)
- Security basic standards follow
- Cost and latency monitor
- Simple system design explain

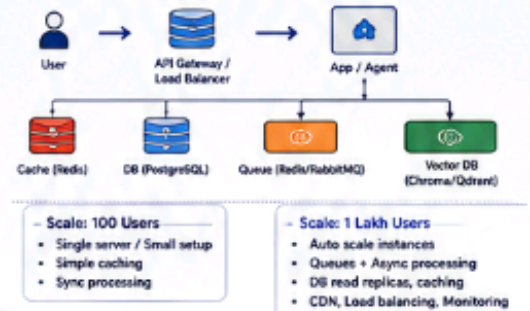


6. COMMON MISTAKES

- ✗ Evaluation skip (output quality not checked)
- ✗ Logs not checked → debug becomes hell
- ✗ Secrets code hardcode
- ✗ Retry without delay → cost waste
- ✗ Fallback not handle → system crash
- ✗ Monitoring not setup → issue late
- ✗ Security ignore → data leak risk



7. SYSTEM DESIGN BASICS (JUST ENOUGH)



8. DEPENDENCY FLOW (Why This Level Comes After Level 7)



9. WEEKLY STRATEGY (Example Path)

- Day 1 - Evaluation Basics + Golden Dataset
- Day 2 - LLM-as-Judge + Rubrics
- Day 3 - Langfuse Setup + Tracing
- Day 4 - Prompt Versioning
- Day 5 - Cost & Latency Monitoring
- Day 6 - Retry, Timeout, Rate Limits
- Day 7 - Fallbacks + Model Routing
- Day 8 - Circuit Breaker + DLQ
- Day 9 - Security Basics + Secrets
- Day 10 - Cloud Deploy (Live)
- Day 11 - Logging + Alerts
- Day 12 - System Design Basics

Repeat & Deepen with Projects!

10. MINI PROJECT IDEAS (Extra Practice)

- Production-ready RAG Chatbot
- AI Email Responder (Full Automation)
- AI Lead Qualification System
- Document Q&A with Evaluation Dashboard
- Personal AI Agent (Memory + Tools + Monitoring)

Add: Observability + Evaluation + Security to all projects

11. NEXT LEVEL PREVIEW (LEVEL 9)

- Level 9 Preview: 4 Master Projects, Polish UI, GitHub + README, Resume, Mock Interviews, DSA, Job Ready & Apply!

Golden Rule: Measure → Monitor → Secure → Scale → Improve | Build Systems That Last. | Think Like an Engineer. Ship Like a Pro.

Code quality matters. Production-ready systems are built with projects.

ONE STEP AT A TIME. MASTER IT.

LEVEL 9

PORTFOLIO + POSITIONING

Show Your Work. Get Hired. Build Your Brand.

Portfolio GitHub Resume LinkedIn Personal Brand Networking Interviews

LEVEL GOAL



तुमचे काम जगाला दिसले पाहिजे.
Strong Portfolio + Personal Brand बना,
ज्यामुळे तुम्हाला चांगल्या संधी मिळतील.



DURATION

No fixed time
(Skill-based)



WEEKLY HOURS

6 - 10 Hours
(Flexible)



DIFFICULTY

Medium
(Consistency Heavy)



IMPORTANCE

Very High
(Career Level)



NEXT LEVEL

Level 10
Senior Skills & Leadership

1. WHAT YOU WILL DO

- Strong Portfolio बना 4-6 High Quality Projects (Real Problems, Real Impact)
- GitHub Optimize करा Clean Code, Proper README, Structure, Docs
- Resume ATS Friendly Skills, Projects, Impact, Metrics
- LinkedIn Optimize Headline, About, Featured, Posts
- Get Interviews Apply Smartly, Referrals, Networking
- Personal Brand Build Content, Learn in Public, Share Knowledge
- Soft Skills Improve Communication, Writing, Presentation
- GitHub Stats & Activity Consistent Commits, Contributions
- Open Source (Optional) Contribute, Collaborate, Build Reputation
- Freelance / Internships First Real World Experience
- Mock Interviews DSA Basics, System Design, Behavioral
- Job Hunt System Track, Apply, Follow-up, Improve

2. TOPICS TO MASTER

- Portfolio Strategy
- Project Selection & Scope
- Problem Statement Writing
- System Design (High Level)
- Clean Code & Best Practices
- README That Converts
- Demo / Screenshots / GIFs
- Deployment & Live Links
- Git & GitHub Mastery
- Resume Building
- LinkedIn Branding
- Writing Good Posts
- Networking (Online/Offline)
- Interview Preparation
- DSA (Basic Level)
- System Design (Basic)
- Behavioral Interviews

PORTFOLIO MUST HAVE



3. PROJECTS TO INCLUDE

- AI Customer Support Agent RAG, Tools, Memory, Multi-channel (Email, Slack)
- AI Lead Automation System n8n + AI Agent, Email, CRM, Scoring, Follow-ups
- Personal AI Knowledge Agent Upload Docs, Ask Anything, Citations, Chat History
- AI Data Analyst Agent Upload CSV, Ask Questions, Charts, Insights
- Multi-Tool AI Agent Platform Agents, Tools, MCP, Workflows, User Management
- Bonus Ideas
 - AI Meeting Notes Bot
 - Content Generator Agent
 - Invoice / Document Automation

WHAT RECRUITERS LOOK FOR?

- Real Projects
- Problem Solving
- Clean Code
- Communication
- Impact / Results

4. TOOLS & TECHNOLOGIES

- Git & GitHub
- VS Code
- Markdown
- Canva / Figma (Designs)
- Vercel / Render
- LinkedIn
- Google Docs / Sheets
- Notion (Documentation)
- Loom (Recording)
- ChatGPT / Claude
- Excalidraw (Diagrams)
- Grammarly (Writing)
- Huntr / Notion Job Tracker
- Mail Tracker (Follow-up)
- Google Analytics (Traffic)

5. CHECKPOINT (Before Level 10)

- 4-6 Projects Deploy करना पूर्ण केले.
- GitHub Clean, Organized आणि Active आहे.
- README Professional आणि Complete आहे.
- Live Demo Links काम करत आहेत.
- Resume ATS Friendly आणि Metrics सहित आहे.
- LinkedIn Profile Optimize केले आहे.
- 15-20 Quality Applications केले आहेत.
- Mock Interviews किमान 5 दिल्या आहेत.
- DSA Basic प्रश्न सोडवले आहेत.
- अपना Personal Brand सुरु केला आहे.



6. COMMON MISTAKES

- Low Quality / Tutorial Copy Projects बनवणे.
- README लिखाण नाही किंवा Poor लिहिणे.
- Code Messy आणि Hard to Understand.
- Live Demo नाही किंवा Deploy नाही.
- Resume मध्ये Metrics / Impact नाही.
- LinkedIn पर काहीच Update नाही.
- Random Apply करणे (No Strategy).
- Interview Preparation नाही करणे.
- Consistency नाही ठेवणे.

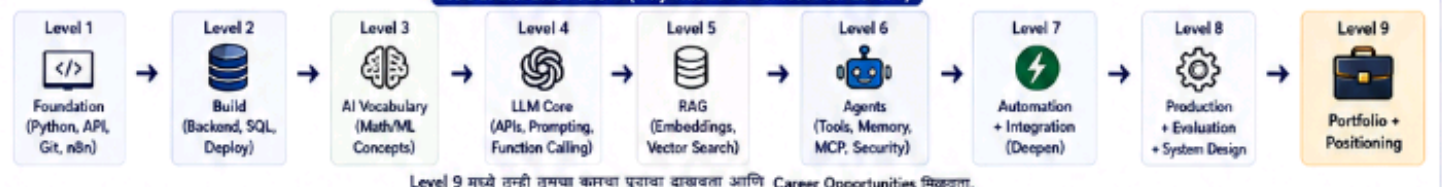


7. PERSONAL BRAND TIPS

- Learn in Public.
- आपले Projects Share करा.
- Tech Topics पर Post / Thread लिहा.
- YouTube / Blog / Twitter पर Content करा.
- इतरांना मदत करा.
- Networking करा (DM, Events, Communities).
- Value द्या, माग सध्या अत्योच्च स्थितीत.



8. DEPENDENCY FLOW (Why This Level Comes After Level 8)



Level 9 मध्ये तुम्ही तुमचा कामचा पुरावा दाखवता आणि Career Opportunities मिळवता.

9. WEEKLY STRATEGY (Example)

- Day 1 - Project Work (Build)
- Day 2 - Code Improve + Docs
- Day 3 - Deploy + Fixes
- Day 4 - README + Screenshots
- Day 5 - LinkedIn / Content
- Day 6 - Applications / Networking
- Day 7 - Mock Interview / DSA

10. MINI PROJECT IDEAS (Extra Practice)

- AI Resume Screener
- AI YouTube Video Summarizer
- WhatsApp / Telegram AI Bot
- AI Voice Notes Transcriber
- AI Expense Tracker

11. NEXT LEVEL PREVIEW (LEVEL 10)

- Senior AI Engineer Skills
- Advanced System Design
- Large Scale AI Systems
- Leadership & Mentoring
- Global Opportunities (Remote / Abroad)
- Higher Salary, Better Impact

YOU WILL GET

- Better Projects
- More Interviews
- More Offers
- Better Career
- Better Life



REMEMBER:

Build → Ship → Share → Improve → Repeat

Consistency + Quality + Patience = Success



Mantra: Build in Public • Solve Real Problems • Help Others • Never Stop Learning



LEVEL 10

SENIOR SKILLS + LEADERSHIP

Scale Impact. Lead Systems. Drive Value. 🚀

🏠 Architecture 👤 Leadership 🎓 Mentorship 💰 Business 💬 Communication

LEVEL GOAL



Become a Senior AI Engineer who can architect, lead, mentor and drive high-impact AI solutions for business growth and innovation.



DURATION

No fixed time
(Skill-based)



WEEKLY HOURS

6 - 12 Hours
(Flexible)



DIFFICULTY

Medium to Hard
(Growth & Leadership)



IMPORTANCE

Extremely High
(Career Defining)



NEXT LEVEL

Level 11
Mastery +
Global Impact

1. WHAT YOU WILL MASTER

- ✓ **Advanced System Design**
Design large scale, event-driven, AI-powered systems
- ✓ **Scalability & Performance**
Scale apps to 1M+ users, high throughput & low latency
- ✓ **Cost Optimization**
Reduce cloud cost, right sizing, auto scaling, reserved instances
- ✓ **Advanced Cloud Architecture**
Multi-region, Multi-account, DR, High Availability
- ✓ **Advanced AI Architectures**
Multi-Agent Systems, Graph RAG, Tool Orchestration
- ✓ **Advanced Data Engineering**
ETL/ELT, Data Lakehouse, Streaming, Real-time pipelines
- ✓ **Security & Compliance**
Zero Trust, IAM, Encryption, Audit, SOC2, GDPR basics
- ✓ **Team Leadership**
Lead projects, plan, delegate, remove blockers
- ✓ **Mentorship & Hiring**
Mentor juniors, conduct technical interviews
- ✓ **Product Thinking**
Understand users, solve real problems, measure impact
- ✓ **Stakeholder Communication**
Present to non-tech, write PRD, reports, demos
- ✓ **Business & Strategy**
ROI, OKR, Roadmaps, Prioritize, Align with business goals
- ✓ **Continuous Learning**
Stay updated, research, read, experiment

2. TOPICS TO MASTER

- Advanced System Design Patterns
- Event Driven Architecture (EDA)
- Microservices vs Monoliths
- API Design (REST, gRPC, GraphQL)
- Caching Strategies (Redis, CDN)
- Database Scaling (Sharding, Replication)
- Message Queues (Kafka, RabbitMQ)
- Container Orchestration (K8s Basics)
- Infrastructure as Code (Terraform)
- CI/CD Advanced (GitHub Actions)
- Observability 2.0 (Logs, Metrics, Traces)
- SRE Basics (SLI, SLO, Error Budgets)
- Advanced AI Architectures (Multi-Agent, Graph, RAG 2.0)
- Data Pipelines at Scale
- Feature Stores & ML Lifecycle
- Cost Governance & FinOps
- Security Best Practices (Advanced)
- Team Management & Leadership
- Technical Roadmaps & Planning
- Communication & Storytelling

SYSTEM DESIGN PILLARS



3. PROJECTS TO LEAD / BUILD

- 🏢 **Enterprise AI Platform**
Multi-tenant AI platform with RAG, Agents, Analytics, Billing, RBAC, Audit Logs
- 📈 **Real-time AI Analytics System**
Stream data → Process → AI Insights → Alerts → Dashboard
- 🤖 **AI Copilot for Business**
Docs Q&A, Task Automation, Report Generation, Integrations
- 💾 **Data Platform (Lakehouse)**
Ingest → Process → Store → Analyze → Visualize
- 🔄 **MLOps End-to-End Pipeline**
Train → Deploy → Monitor → Retrain → Versioning

PROJECT SUCCESS CHECKLIST

- ✓ Solves real world problem
- ✓ Scalable, Secure, Reliable
- ✓ Well documented
- ✓ Deployed & live
- ✓ Monitored & improved
- ✓ Used by real users

4. TOOLS & TECHNOLOGIES

- ☁️ Cloud: AWS / GCP / Azure
- 🐳 Containers: Docker, Kubernetes
- 🔧 IaC: Terraform, CloudFormation
- 🔄 CI/CD: GitHub Actions, ArgoCD
- 🗄️ Databases: PostgreSQL, MySQL, MongoDB, BigQuery
- 📡 Streaming: Kafka, Redis Streams
- 🔥 Monitoring: Prometheus, Grafana, ELK, Datadog
- 🧠 AI: OpenAI, Anthropic, LangChain, LangGraph, Vector DBs
- 📦 Vector DB: Pinecone, Qdrant, Weaviate
- 🔄 Workflow: n8n, Temporal, Airflow
- 🛡️ Security: Vault, IAM, WAF, KMS
- 🌐 Other: Git, Notion, Slack, Figma, Miro, Postman, VS Code

MINDSET

- 🧠 Think Architect
- 👤 Act Leader
- 👥 Mentor Others
- 🌟 Create Impact

5. CHECKPOINT (Before Moving to Level 11)

- ✓ Led & shipped 3+ high impact projects
- ✓ Designed scalable, secure, cost optimized systems
- ✓ Handled production systems with monitoring
- ✓ Mentored juniors & conducted interviews
- ✓ Strong portfolio with real world impact
- ✓ Can explain system design end-to-end
- ✓ Comfortable with stakeholders & presentations
- ✓ Understand business, ROI & user needs



6. COMMON MISTAKES TO AVOID

- ✗ Over-engineering solutions
- ✗ Ignoring monitoring & alerts
- ✗ Not thinking about cost
- ✗ Poor documentation
- ✗ Not validating with real users
- ✗ Micromanaging team
- ✗ Ignoring security
- ✗ Stop learning after success



7. LEADERSHIP & GROWTH

- ★ Lead by Example
- ★ Take Ownership
- ★ Empower Team
- ★ Give Feedback
- ★ Be a Problem Solver
- ★ Build Trust
- ★ Communicate Clearly
- ★ Think Long Term



8. DEPENDENCY FLOW (Why This Level Comes After Level 9)



9. WEEKLY STRATEGY (Example)

- Day 1: Learn / Deep Work (2-3 hrs)
 - Day 2: Project Work (2-3 hrs)
 - Day 3: System Design Study (1-2 hrs)
 - Day 4: Review + Improve (1-2 hrs)
 - Day 5: Team / Mentorship (1 hr)
 - Day 6: Read / Research (1-2 hrs)
 - Day 7: Rest / Reflect / Plan Next Week
- Consistency > Motivation

10. MINI PROJECT IDEAS (Extra Practice)

- 🔍 AI Resume Screener & Ranker
- 🧠 Multi-Agent Research Assistant
- 🛡️ Real-time Fraud Detection System
- 📄 Intelligent Document Processing Platform
- 🔗 Custom GPT + RAG for Your Domain
- 💰 Personal Finance AI Advisor
- 🗣️ AI Powered Meeting Assistant
- 🔧 IoT + AI Predictive Maintenance

11. NEXT LEVEL PREVIEW (LEVEL 11)

- Enterprise Architecture
 - Advanced ML / Deep Learning
 - AI Safety & Alignment
 - Global Scale Systems
 - Product Management
 - People Management (Advanced)
 - Thought Leadership & Speaking
 - Global Opportunities / Remote Leadership
- Level 11 is about Mastery & Global Impact!

YOU WILL GET

- ✓ High Paying Career
- ✓ Global Opportunities
- ✓ Leadership Position
- ✓ Financial Freedom
- ✓ Impact & Recognition
- ✓ Ability to Build Products that Change Lives



★ **KEY MANTRA:** Learn → Build → Ship → Lead → Mentor → Impact | Stay Consistent. Keep Building. The World Needs Builders Like You! 🚀